SOUTENANCE OCR

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SOMMAIRE

- Architecture du projet
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ARCHITECTURE DU PROJET

Approche

- Architecture modulaire bas niveau en C
- Séparation claire : fichiers .c / .h
- Pipeline linéaire et découplé

Pipeline global

Image brute
→ Prétraitement
→ Segmentation
→ OCR
→ Reconstruction
→ Solver
→ Affichage

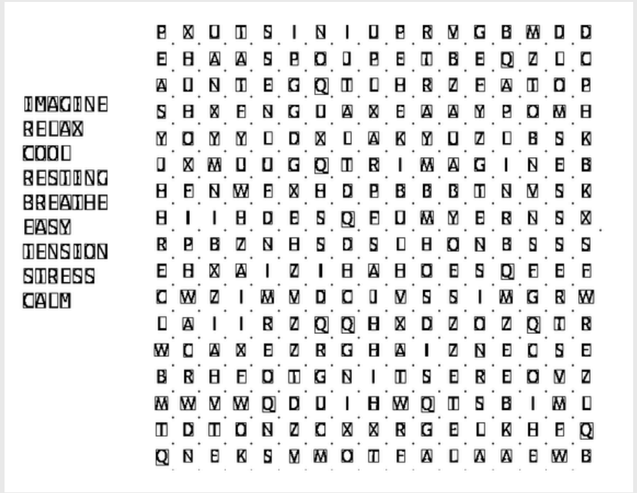
Avantages

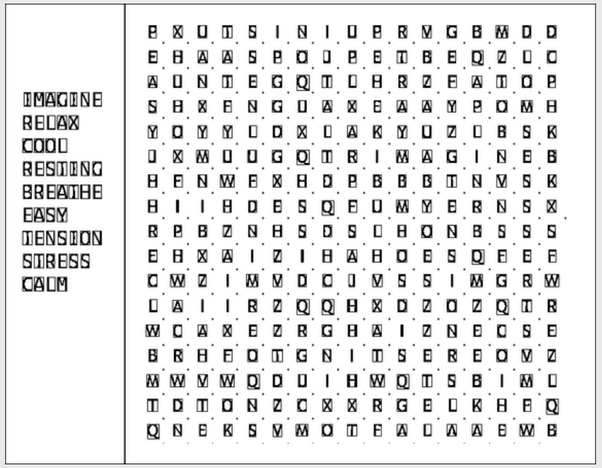
- Débogage facilité
- Robustesse globale
- Modules testables indépendamment
- Travail en parallèle possible

PRÉTRAITEMENT DE L'IMAGE



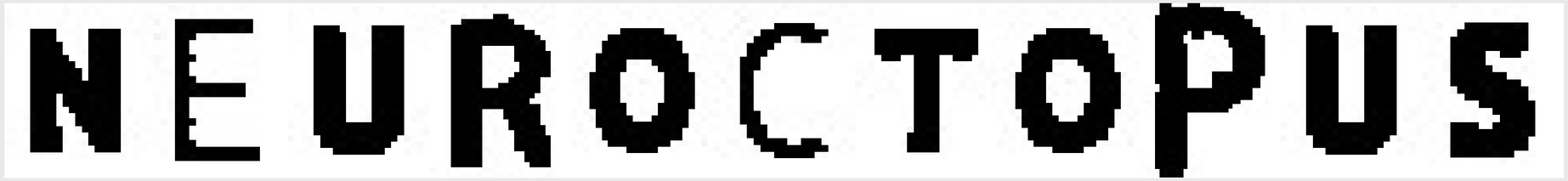






RÉSEAU DE NEURONES

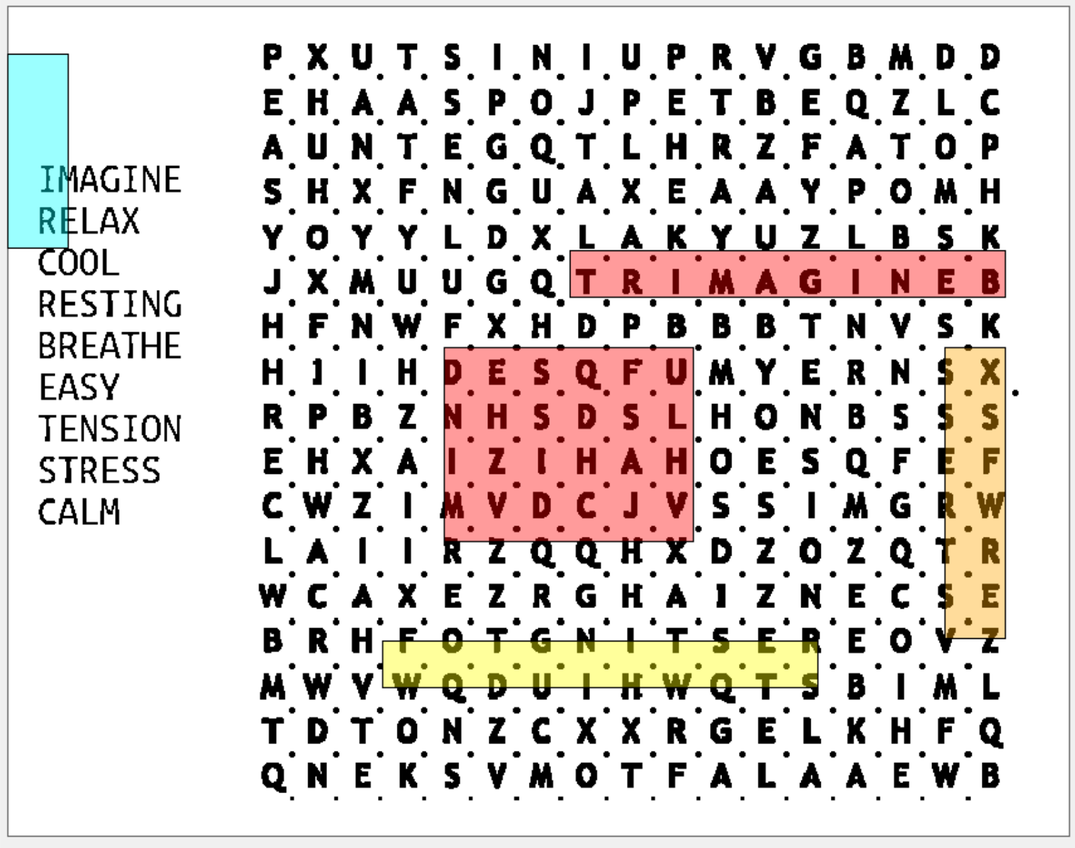
- Images d’entrée : 32×32 pixels
- Couche cachée : 256 neurones
- Base d’entraînement : 170 images
- Entraînement : ~10 000 époques



RECONSTRUCTION LOGIQUE

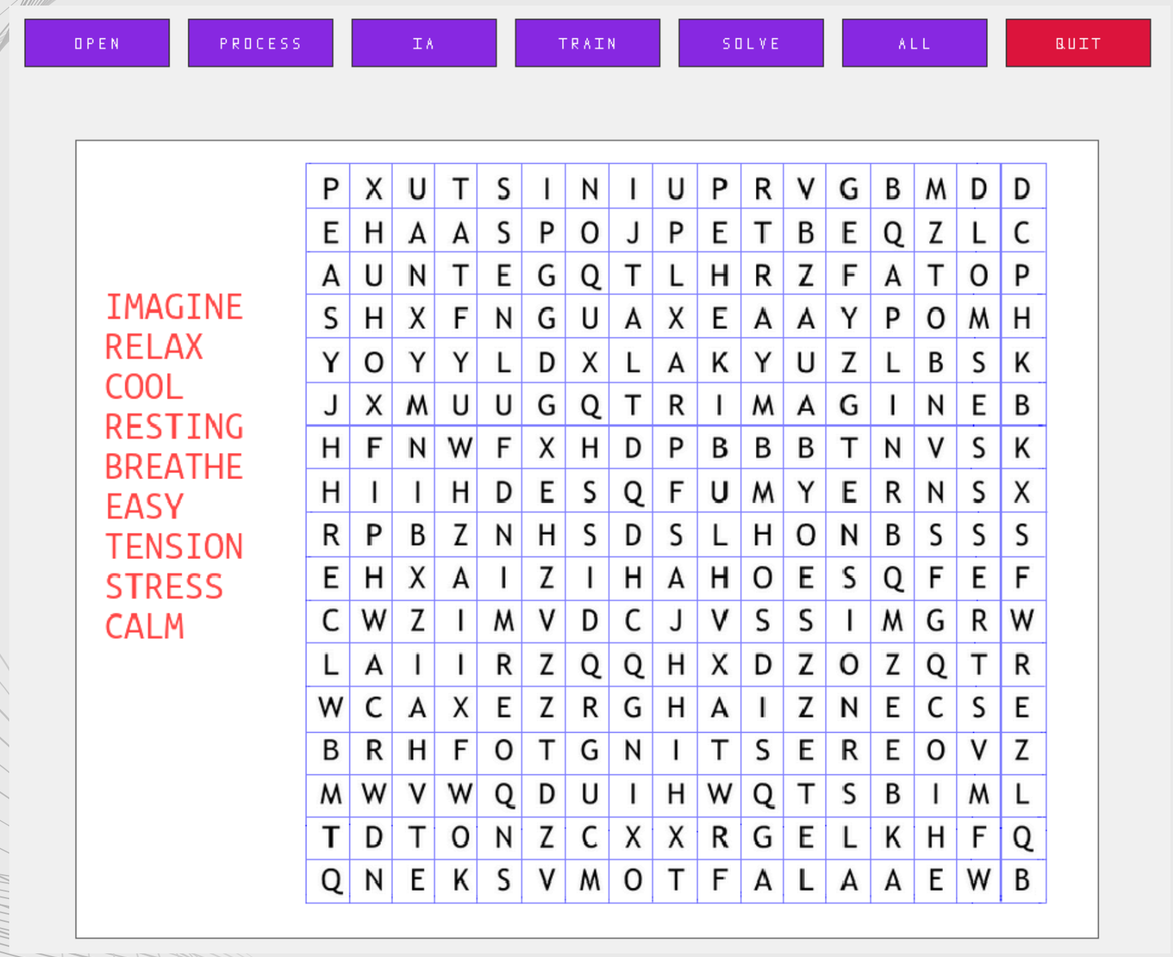
```
Word to find: 'IMAGINE'
Word to find: 'RELAX'
Word to find: 'COOL'
Word to find: 'RESTING'
Word to find: 'BREATHE'
Word to find: 'EASY'
Word to find: 'TENSION'
Word to find: 'STRESS'
Word to find: 'CALM'
```

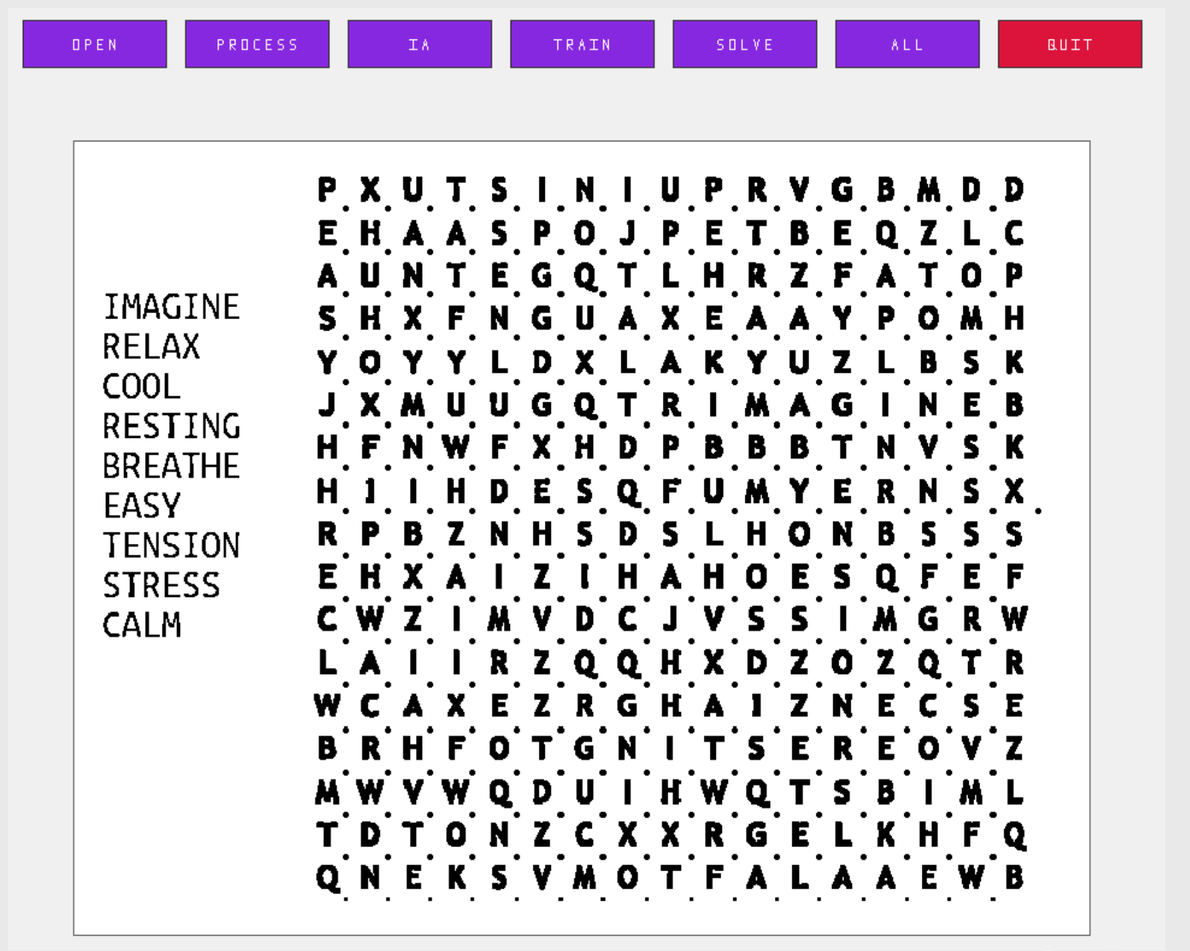
```
 0: PXUTSINIUPRVGBMDD
 1: EHAASPOJPETDEQZLC
 2: AUNTEGQTLHRZFATGP
 3: SHXFNGUAXEAAYPOMH
 4: YOYYLDRLAKYUZLBSK
 5: JXMUUGQTRIMAGINED
 6: HFNWEXHDPBSDTNVSK
 7: HIIHDESQPUMYERNSX
 8: RPBZNHSDSLNONBSSS
 9: EHRAIZIHAHOESQFEE
10: CWZIMVDCJVSSIMGRW
11: LAIIRZQQHXDZGZQTR
12: WCAXEZRGHAIZNECSE
13: DRHFOTGNITSEREOVZ
14: MWVWQDUIHWQTSBIML
15: TDTONZCXXRGELKHEQ
16: QNEKSVMOTFALAAEWB
```




```
Found word: IMAGINE at (5,9) to (5,15)
Found word 'IMAGINE' at: (5, 9) to (5, 15)
Word 'RELAX' not found
Word 'COOL' not found
Found word: RESTING at (13,12) to (13,6)
Found word 'RESTING' at: (13, 12) to (13, 6)
Word 'BREATHE' not found
Found word: EASY at (1,0) to (4,0)
Found word 'EASY' at: (1, 0) to (4, 0)
Word 'TENSION' not found
Found word: STRESS at (12,15) to (7,15)
Found word 'STRESS' at: (12, 15) to (7, 15)
Found word: CALM at (10,7) to (7,10)
Found word 'CALM' at: (10, 7) to (7, 10)
Found 5/9 words. Drawing highlights...
Solution from 1: (5,9) to (5,15)
Solution from 4: (13,12) to (13,6)
```

INTERFACE GRAPHIQUE





ANALYSE CRITIQUE

LIMITES

- La Rotation
- Images avec defauts
- Le resultat en couleurs

AMELIORATIONS

- La Rotation
- Amélioration du réseau de neurones
- Le résultat en couleurs

CONCLUSION